

Class 16: Subspaces

Definition: A subspace of $\mathbf{R}^n$ is any set H of vectors in $\mathbf{R}^n$ having the following three properties

1. H is not an empty set.
2. H is closed under vector addition.
3. H is closed under scalar multiplication.

Examples: Let A be any $m \times n$ matrix.

1. The set of all solutions to the homogeneous system $A\mathbf{x} = \mathbf{0}$ is a subspace of $\mathbf{R}^n$ called the null space of A .

Proof

2. The set of all linear combinations of the columns of A is a subspace of $\mathbf{R}$ called the column space of A .

Proof: Let $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ be the columns of the matrix A .

Definition: A basis for a subspace H is a set of vectors in H that spans H and is linearly independent.

Examples:

1. The set $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is the standard basis for $\mathbf{R}^n$.
2. Any set of n linearly independent vectors in $\mathbf{R}^n$ will be a basis for $\mathbf{R}^n$ since any matrix with these vectors as columns will have n pivot columns and thus n pivot rows so will be a spanning set for $\mathbf{R}^n$ as well.

Column Space and Null Space Practice

The matrix $A = \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 3 & 12 & 1 & 5 & 5 \\ 2 & 8 & 1 & 3 & 2 \\ 5 & 20 & 2 & 8 & 8 \end{bmatrix}$ reduces to $\begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

a) Find a basis for the column space of A.

b) Is the vector $\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ in the column space of A?

c) Find a basis for the null space of A.

d) Find a basis for the row space of A